

Website Analytics Dashboard

Implementation playbook for the OMGEXP admin **/analytics** page — reusable on other sites.
Reference stack: Next.js + GA4 Data API + Search Console API + GTM on the public marketing site.
Version: July 2026

1. Architecture

The dashboard is **not** an embedded Google Analytics UI. It is a custom admin page that pulls aggregated metrics from the **GA4 Data API** using a Google Cloud service account, then renders charts with Recharts.

Search → Google results (Search Console logs impressions/clicks) → Visitor clicks → Marketing site (GTM) → GA4 session

Staff → /analytics → GET /api/analytics/ga → ga-data.ts (GA4) + gsc-data.ts (Search Console) → charts

Contact form → attribution in localStorage → Supabase (CRM) – separate from this dashboard

Two halves: (A) code in your repo embeds GTM and pushes events; (B) configuration in GTM UI + GA4 Admin (tags, triggers, custom dimensions, referral exclusions) is **not** in git.

2. Code-side setup (portable)

2.1 Environment variables (Tr backend)

Variable	Required	Description
GA4_PROPERTY_ID	Yes	Numeric property ID (GA4 Admin → Property settings)
GA_SERVICE_ACCOUNT_EMAIL	Yes	Service account email from Google Cloud
GA_SERVICE_ACCOUNT_PRIVATE_KEY	Yes	PEM key; use \n for newlines in env
MARKETING_GA_HOSTNAME	No	Default <code>www.omgcargo.tech</code> — filters event-scoped reports
GA_PUBLIC_PATH_PREFIX	No	Default <code>/site</code> — legacy marketing paths on the staff app domain
GSC_SITE_URL	No	Search Console property URL (must match Console exactly)
GSC_LAG_DAYS	No	Default 3 — end date offset for Search Console queries
NEXT_PUBLIC_GTM_ID	No	GTM on public pages (separate from dashboard API)

2.2 Key source files

- `src/lib/ga-range.ts` — GA4 relative dates + Search Console absolute dates
- `src/lib/ga-data.ts` — GA4 Data API queries, site filter, lead events
- `src/lib/gsc-data.ts` — Search Console API (impressions, clicks, queries, opportunities)
- `src/lib/gsc-opportunities.ts` — striking distance, low CTR, zero-click ranking
- `src/app/api/analytics/ga/route.ts` — admin-only API + 30-minute in-memory cache
- `src/app/(main)/analytics/page.tsx` — dashboard UI

2.3 Site filter logic

Reports include traffic where **either**:

- `hostName` = `MARKETING_GA_HOSTNAME` (new Astro site at `www.omgcargo.tech`), or
- `pagePath` begins with `GA_PUBLIC_PATH_PREFIX` (legacy `/site/*` on `cargo.omgexp.com`)

Staff-only routes (e.g. `/dashboard`, `/login`) are excluded unless they match the above.

2.4 Marketing site events (dataLayer)

Push these from the public site after GTM loads:

Event	Parameters	When
<code>generate_lead</code>	<code>form_name</code> , optional <code>transaction_id</code>	After contact API returns success (not on click)
<code>cta_click</code>	<code>cta_label</code> , <code>cta_location</code>	Primary CTA clicks
<code>contact_click</code>	<code>contact_method</code> , <code>contact_value</code>	tel: / mailto: links

3. Manual setup (GTM + GA4 Admin)

1. Create a GA4 property and web data stream for your marketing domain.
2. Create a GTM container; add GA4 Configuration tag with your Measurement ID (G-xxxx).
3. Add Custom Event triggers for `generate_lead`, `cta_click`, `contact_click`.
4. Add GA4 Event tags mapping `dataLayer` parameters to event parameters.
5. Optional: register event-scoped dimension **form_name** in GA4 Admin only if you have multiple forms and want form-level breakdown elsewhere — the dashboard uses `pagePath` for “Leads by page” and needs no custom dimension.
6. Enable **Google Analytics Data API** in Google Cloud; create a service account; download JSON key or copy email + private key to env.
7. GA4 Admin → Property access management → add service account as **Viewer**.
8. GA4 Admin → Data streams → Configure tag settings → define **unwanted referrals** (e.g. your own domain) to fix self-referral like `omgcargo.tech / referral`.
9. Mark `generate_lead` as a key event in GA4 if you want it in standard reports.

3b. Manual setup (Search Console)

1. Verify your site in Google Search Console (domain or URL-prefix property).

2. Google Cloud → enable **Search Console API** (same project as GA4 service account).
3. Search Console → Settings → Users and permissions → add service account email as **Restricted**.
4. Set GSC_SITE_URL to the exact property string: sc-domain:example.com Or <https://www.example.com/>.
5. Optional: adjust GSC_LAG_DAYS (default 3) if reports look incomplete for recent days.

4. What each dashboard metric means

UI label	Source	Means	Does NOT mean
Search impressions	Search Console	Times your site appeared in Google results	GA4 page views
Search clicks	Search Console	Clicks from Google search to your site	All sessions (includes direct, ads, social)
Search CTR	Search Console	Clicks ÷ impressions	Conversion rate
Avg position	Search Console	Weighted average rank in results	Exact rank for every query
Users / Sessions / Page views	GA4 filtered totals	Marketing + legacy /site paths	All staff app usage
Lead CVR	GA4 computed	Lead events ÷ sessions	Revenue or CRM close rate
Lead events	GA4 generate_lead	Form submits tracked in GA4	Unique people; phone clicks
Leads by page	GA4 pagePath + eventCount	Page where the lead event fired	First page in session (see Top entry pages)
Micro-conversions	GA4 events	generate_lead, cta_click, contact_click	Deduplicated users
Top entry pages	GA4 landingPage + sessions	First page in each session	Most viewed pages (use Top pages)
Channel performance	GA4 merged table	Sessions + leads + CVR by source/medium	Multi-touch attribution
Top sources	GA4 sessionSourceMedium	Source/medium per session	Search Console queries

Why GSC clicks ≠ GA4 organic sessions: GSC counts Google-search clicks only (server-side). GA4 counts all sessions that reach your site (client-side), grouped by last-click source/medium. Ad blockers suppress GA4 but not GSC. Direct visits and non-Google channels appear in GA4 only.

Period-over-period % compares equal-length windows. GA4 ends **yesterday**; Search Console ends **today** – **GSC_LAG_DAYS**.

4b. SEO opportunities card (Search Console)

Uses up to 250 query/page rows from the same GSC API calls (no extra quota). Three columns:

Column	Filter	What to do next
Striking distance	Queries ranking #5–20 with ≥ 10 impressions	Add content or internal links targeting these terms — small ranking gains can move you to page 1
Low CTR pages	Pages with CTR below site average, sorted by <i>missed clicks</i>	Rewrite title and meta description so the snippet matches search intent
Zero-click queries	Impressions > 0 but zero clicks	Snippet or intent mismatch — consider a dedicated landing page or clearer title

5. Accuracy limits

- **Processing lag:** Standard reports lag 24–48 hours; Realtime/DebugView are immediate.
- **Undercount:** Ad blockers, no-JS, Safari ITP — typically 10–30% of sessions missing client-side.
- **Lead events vs CRM:** GA4 counts events; Supabase stores submissions with attribution JSON — two systems, two numbers.
- **Double submits:** eventCount increments per event; GA4 dedupes transaction_id only for purchase, not generate_lead.
- **Legacy /site traffic:** If old marketing routes stay live, metrics mix two sites until you redirect or narrow filters.
- **Self-referral:** Cross-domain hops without referral exclusion inflate “referral” and split sessions.

End-to-end verification not yet done: Submit one real test lead via the live contact form, then confirm (1) GA4 DebugView shows generate_lead, (2) dashboard Lead events increments after lag, (3) Supabase contact_submissions.attribution has UTM/click IDs.

6. Verification checklist

1. GTM Preview → submit test form → see generate_lead with form_name.
2. GA4 DebugView → same event within seconds.
3. GA4 Realtime → active user on marketing hostname.
4. Admin /analytics → KPIs load (not “Analytics not configured”).
5. If a section shows **Data unavailable** with an error string, fix GA4 or Search Console config — lead breakdown uses standard pagePath, not custom dimensions.
6. Search Console section hidden until GSC_SITE_URL is set and API access works.
7. Run node --experimental-strip-types scripts/ga-range.test.mjs after changing date logic.

8. Compare lead events (GA4) vs contact rows (Supabase) over the same week — expect difference, not parity.

7. Appendix

7.1 Env template

```
GA4_PROPERTY_ID=123456789
GA_SERVICE_ACCOUNT_EMAIL=analytics-read@project.iam.gserviceaccount.com
GA_SERVICE_ACCOUNT_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\n...\n-----END PRIVATE KEY-----\n"
MARKETING_GA_HOSTNAME=www.example.com
GA_PUBLIC_PATH_PREFIX=/site
GSC_SITE_URL=sc-domain:example.com
GSC_LAG_DAYS=3
NEXT_PUBLIC_GTM_ID=GTM-XXXXXXX
```

7.2 PEM key troubleshooting

If the API returns `OpenSSL DECODER routines::unsupported`, the private key env is malformed. The codebase normalizes quotes, literal `\n`, and Windows `\r` — but the key must still be a valid PEM block.

7.3 Legacy /site vs new domain

OMG currently serves marketing on both `www.omgcargo.tech` (Astro) and `/site/*` on the Next app. Top pages may show `/site/login` alongside `/`. Options: 301 legacy to new domain (SEO + cleaner analytics), or set `GA_PUBLIC_PATH_PREFIX` to a path that no longer receives traffic once migrated.

7.4 PDPA note

GTM/GA4 cookies require disclosure in your cookie policy. Attribution keys in `localStorage` should be listed there too. Consent Mode v2 is recommended for new deployments in regulated markets.